Usage Models

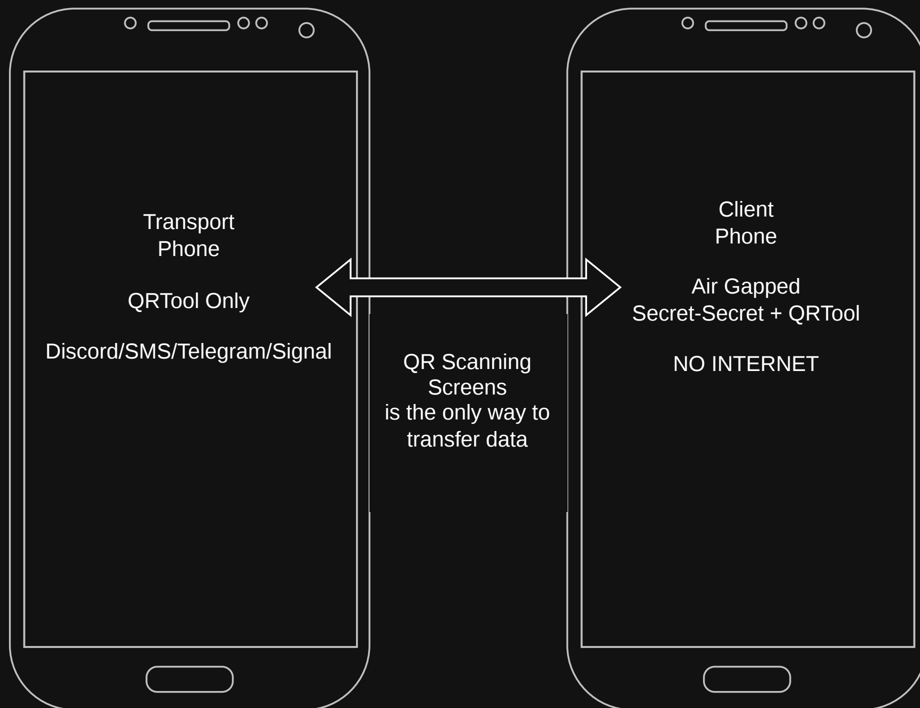
QRTool + Secret-Secret

This the model the Software is built around. Each app requires at least Android 10 (2019 +)

The Transport Phone.
This can be your normal everyday driver device. Normal Access to everyday messaging apps.

Client Phone.
The client phone should ideally never come online. As the Secret-Secret App keeps all keys in plain text to make the decode of messages easy. The offline phone does not need cell service and should be in airplane mode at all times.

Messages are sent via the camera and QR codes to make the encryption and decryption of data as easy as possible.



QRTool + Secret-Secret + Meshtastic

An Alternativeway to send messages over offline networks. QRTool and Secret-Secret app requires at least Android 10 (2019 +)

Meshtastic app becomes the transport layer.

Client Phone.
The client phone should ideally never come online. As the Secret-Secret App keeps all keys in plain text to make the decode of messages easy. The offline phone does not need cell service and should be in airplane mode at all times.

Messages can be copy pasted to easy message management.

QRTool + Secret-Secret

The model brings the most risk and is not recommend but can be used in a safer manner. Each app requires at least Android 10 (2019 +)

The Transport Phone + Client Phone.
This can be your normal everyday driver device. Normal Access to everyday messaging apps.

In this mode keys and messages are exposed to anybody that has access to your device, local or remote since all data is on this device the assembly of data can be trivial via brute force of scanning all your message apps for base32 strings.

The only way this model works is to not save any messages and use the encrypt/decrypt option. This will not save any messages to the device. Data stays encrypted in your normal message app.
Its also recommend to delete any encrypted messages received, while this will not stop a transport service from keeping a copy of the data at least its not on the device. Delete Contacts, delete messages and roll keys often.

QRTool + Secret-Secret + Meshtastic

The model also brings risk and is not recommend but can be used in a safer manner. Each app requires at least Android 10 (2019 +)

The Transport Phone + Client Phone.
This can be your normal everyday driver device. Normal Access to everyday messaging apps.

Send messages over Meshtastic, while this will help with data privacy as copies will be made local nodes only. It still poses risk at the device level.

In this mode keys and messages are exposed to anybody that has access to your device, local or remote since all data is on this device the assembly of data can be trivial via brute force of scanning all your message apps for base32 strings.

The only way this model works is to not save any messages and use the encrypt/decrypt option only. This will not save any messages to the device. Data stays encrypted in your normal message app.
Its also recommend to delete any encrypted messages received, while this will not stop a transport service from keeping a copy of the data at least its not on the device. Delete Contacts, delete messages and roll keys often.

Setup

QRTool + Secret-Secret

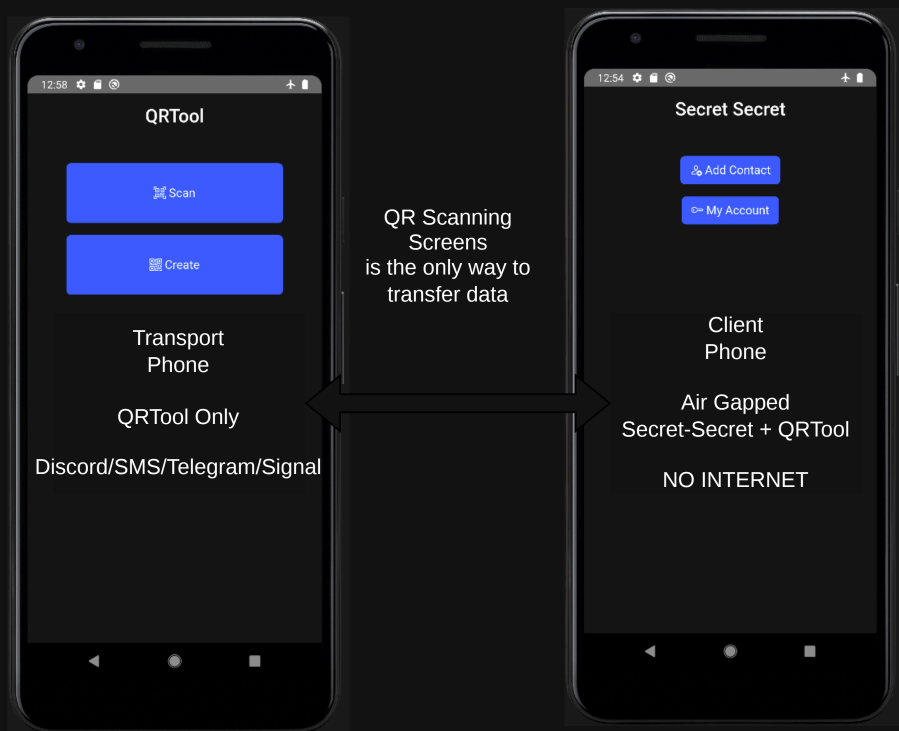
This the model the Software is built around. Each app requires at least Android 10 (2019 +)

We will be covering the basic setup using the recommend setup, two phones.

Start by downloading and installing the apks from the github or download the source code and create your own signed APKs

Transport will only need the QRTool and will not have any key or account data. It will only be used to encode and decode QR codes between devices.

Client Phone Install Secret-Secret.



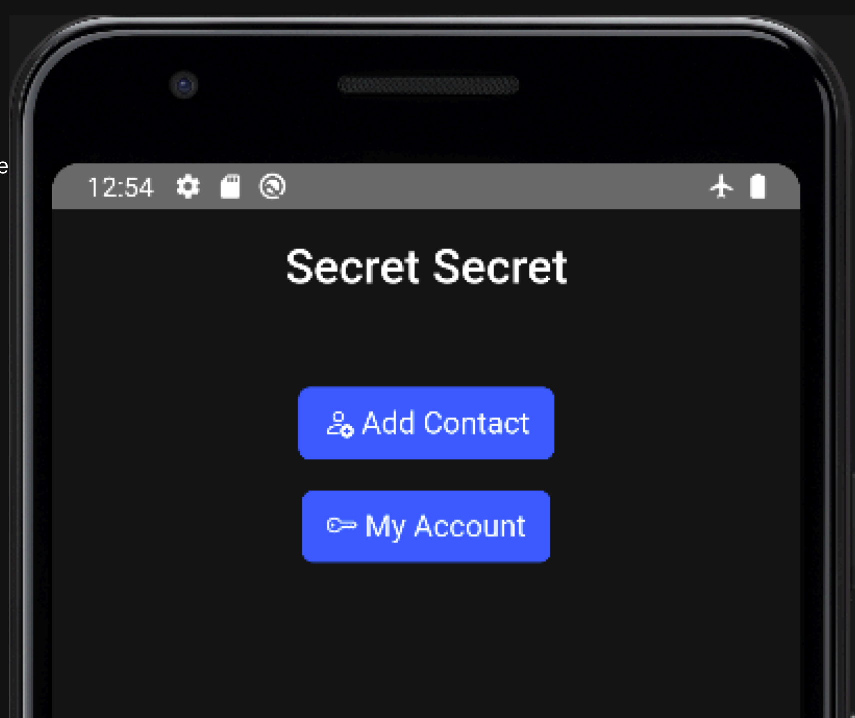
Secret-Secret

We have two buttons on the main screen. Add Contact and My Account.

The Add Contact will allow you to save someone else's key to your device by scanning their QR Code of the 8 word encryption pass phrase.

The 8 word encryption pass phrase uses the same bip-0039 word list used by bitcoin and should be good enough security while still making recovery possible.

Next we will cover how to create your own encryption pass phrase.



Secret-Secret

My Account.
Here is where your 8 word encryption pass phrase is saved for sharing. The box with "pc" is whatever you want to call yourself.

The next box down is where your key is.

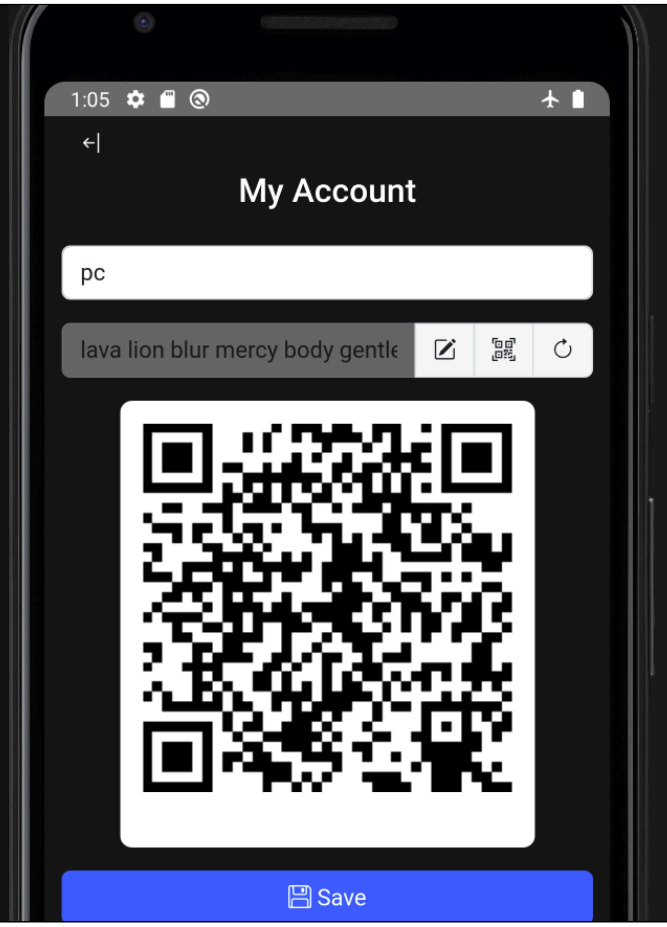
Clicking the refresh icon will create a new random key using the same bip-0039 as stated before. Keys are saved in the database each time you click save but are not viewable. If you roll your key, then someone sends you a message using an old key, it will not decrypt it. This is a design choice to ensure that your contacts always have your new keys.

While we dont force key rotation. If you believe a contact has been compromised, your key will be exposed, you will need to share this new key to your trusted contacts in a form of trusted communications that is not the primary share of encrypted data. In person is best as the qr scan makes this a quick process without manual key entry.

The edit button allows you to manually change your key to whatever you want, small alterations made to your existing key is an effective way to update your keys.

The QR Scan button allows you to scan any qrcode, the update your key with its contents.

No changes to your key will be saved unless your click Save.



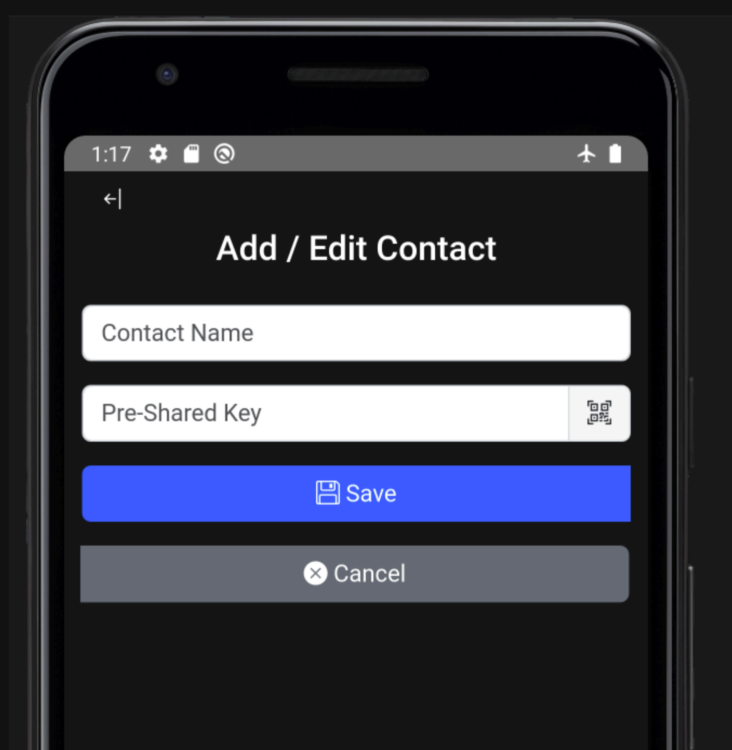
Secret-Secret

Add Contact.

Enter Contact Name

Click the QR Scan Button or type in the contact's key.

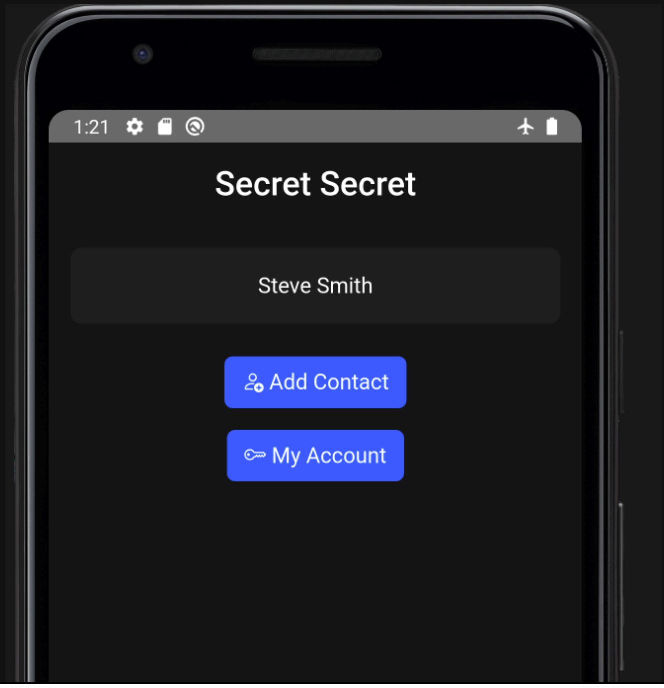
Click Save to save



Secret-Secret

On my main screen after clicking save I have a contact named "Steve Smith" for my example.

Click on Steve to start an exchange.



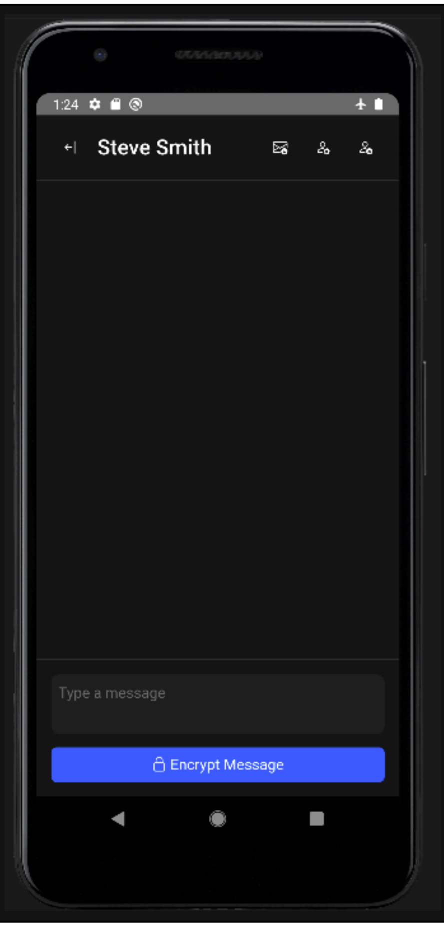
Secret-Secret

On the contact screen we have a few options to cover. Name at the top, next in order.

- Add Message. we will cover this later.
- Edit Contact. Edit Contact, it is the same screen as when we created the contact.
- Delete contact. Removes contact record and messages.

At the bottom of the screen we have the selection to enter a message, do this and then click "Encrypt Message" button.

Review source code if you want a detailed break down. All cyrpto functions will be kept in crypto.js ChatGPT put the time to crack using our default 8 words at 9.8 trillion years to crack brute force even with my lazy design choices, your data is safe as long as the keys are. This is ***critical***.



Secret-Secret

Now we get our message display.

Use the Transport phone to scan the first using QRTool.

Verify the output string matches on each screen. If they do, send away and await a reply.

You have the option to save this message by clicking the "Store" Button. You do not have to do this, and can click the close button, doing this may help your general Operations Security as this is all in memory until you click that "Store" button and create a database write to the local sql lite file. For this Setup I will click "Store".

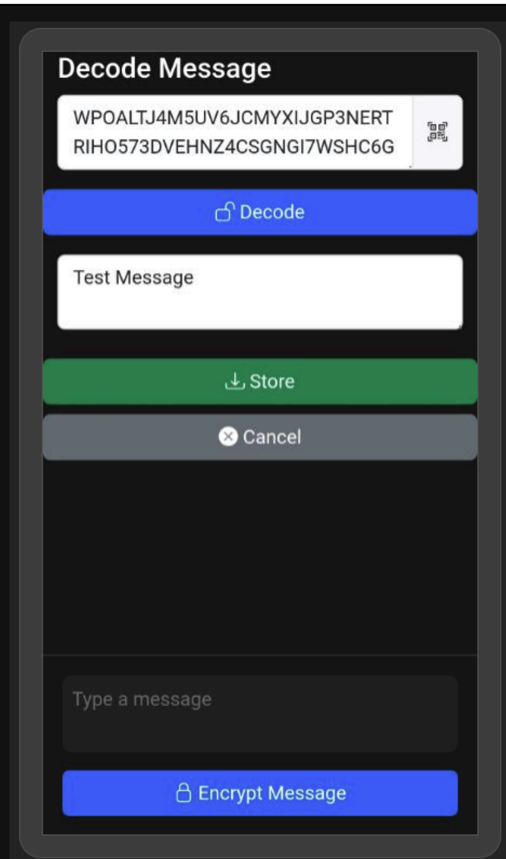
If you chose to store this message the message window will be dismissed and your message will show decrypted in the messages window.

Messages are only decrypted in memory, they are encrypted in the database.

Secret-Secret

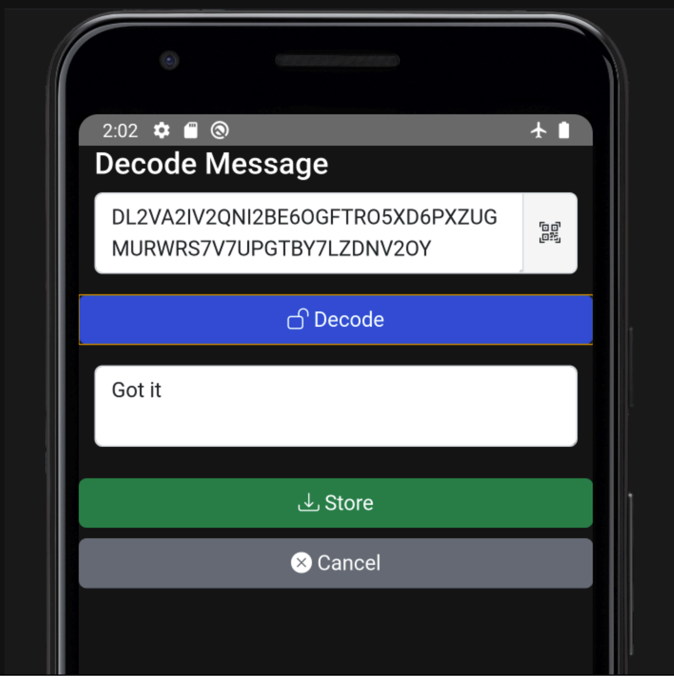
On the contacts transport phone, they will open QRTool.

- Click "Create"
- Paste your encrypted text
- Click "Create"
- Open Secret-Secret on their client device.
- Open your contact.
- Click the Decode Message Button.
- Click the QR Scan button.
- Click Decode
- Make the choice to store this message with the "Store" Button.



Secret-Secret

They send a reply, you do the same as above.
Click Store if you want.



Secret-Secret

That is that all for now. The code is open source, there will be bugs, they will need to be fixed by people like you.
Help.
Thanks -Adamz01h